

Solving by completing the square

WORKSHEET



Interactive version

Name _____

Class _____

Date _____

REMEMBER

Squares

Level 1 • threshold

1.

Which expression is equal to $(x + 3)^2$?

☐ A $x^2 + 9$

☐ B $x^2 + 6x + 9$

☐ C $x^2 + 6x + 6$

☐ D $x^2 + 3x + 9$

Level 1 • threshold

2.

Solve $(x + 5)^2 = 4$.

Answer: _____

Level 1 • threshold

3.

Which number in the box makes $x^2 + 10x + \square$ the square of a binomial?

☐ A 25

☐ B 5

☐ C 10

☐ D 100

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 2 • satisfactory

Solve $x^2 - 12x + 20 = 0$ by completing the square.

Answer: _____

Example 2 • An odd b: $x^2 + 5x + 4 = 0$

MOVE C

$$x^2 + 5x = -4.$$

ADD $(5/2)^2 = 25/4$ TO BOTH SIDES

$$x^2 + 5x + \frac{25}{4} = -4 + \frac{25}{4} = \frac{9}{4}.$$

WRITE THE SQUARE

$$\left(x + \frac{5}{2}\right)^2 = \frac{9}{4}.$$

SOLVE • YOUR TURN

YOUR TURN

Level 2 • satisfactory

In lesson 1.1, Ona tried to solve $x(x - 3) = 4$ and got it wrong. Solve it properly: multiply out, then complete the square.

Answer:

Example 3 • When $a \neq 1$: $4x^2 - 8x + 3 = 0$

DIVIDE BY $A = 4$

$$x^2 - 2x + \frac{3}{4} = 0. \text{ The rule needs the coefficient of } x^2 \text{ to be 1.}$$

MOVE C

$$x^2 - 2x = -\frac{3}{4}.$$

ADD $(-2/2)^2 = 1$ TO BOTH SIDES

$$x^2 - 2x + 1 = \frac{1}{4}, \text{ that is, } (x - 1)^2 = \frac{1}{4}.$$

SOLVE • YOUR TURN

CHECK • YOUR TURN

YOUR TURN

Level 2 • satisfactory

Solve $9x^2 - 18x + 5 = 0$ by completing the square.

Answer:

CHECK

Check yourself

Level 1 · threshold

1.

Which number must you add to $x^2 - 8x$ to make it the square of a binomial?

Answer:

Level 1 · threshold

2.

Which equation is the same as $x^2 - 6x = 16$?

- ☐ A $(x - 3)^2 = 25$ ☐ B $(x - 3)^2 = 16$ ☐ C $(x + 3)^2 = 25$ ☐ D $(x - 6)^2 = 52$

Level 2 · satisfactory

3.

Write $x^2 + 7x + 10$ in the form $(x + p)^2 + q$. Enter p and q , in this order, separated by a semicolon.

Answer:

Level 2 · satisfactory

4.

Solve $x^2 + 3x - 10 = 0$ by completing the square.

Answer:

Level 2 · satisfactory

5.

Complete the square in each equation. Which one has **exactly one** root?

- ☐ A $x^2 + 14x + 40 = 0$ ☐ B $x^2 + 49 = 0$ ☐ C $x^2 + 14x + 49 = 0$ ☐ D $x^2 + 14x + 50 = 0$

Level 2 · satisfactory

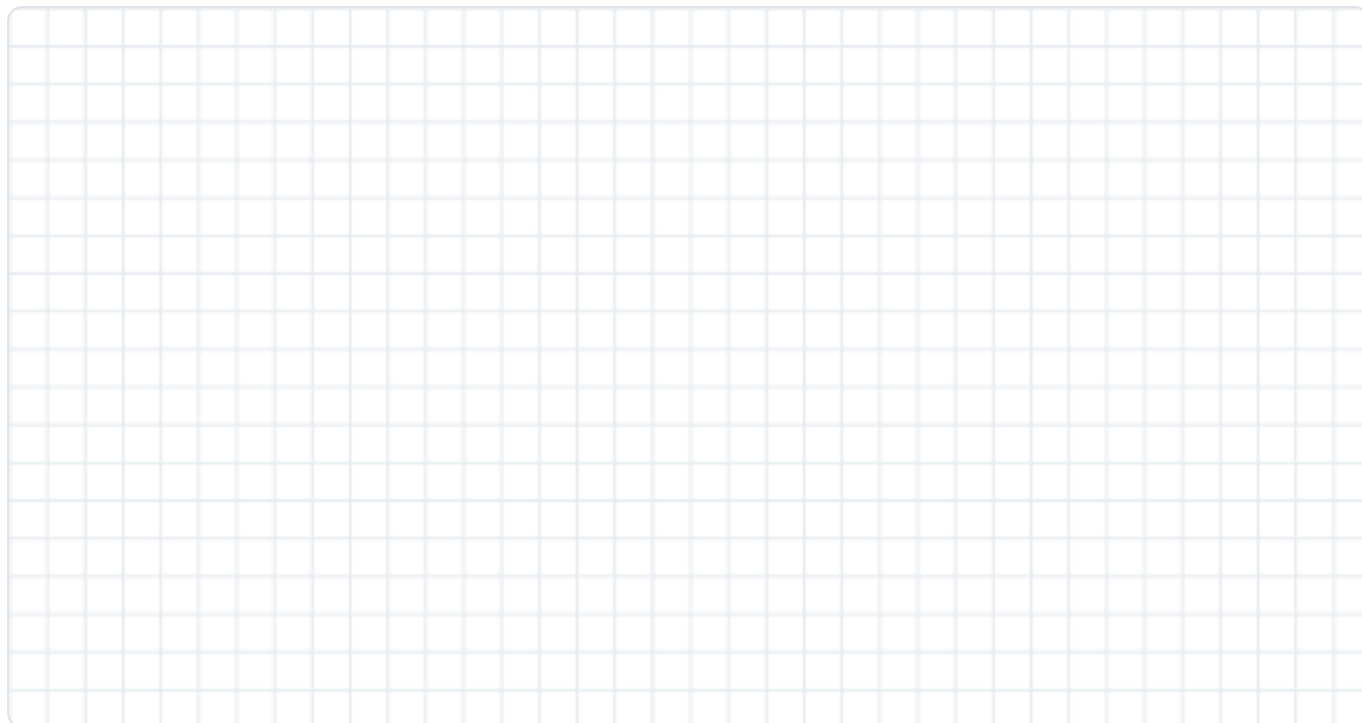
6.

Put the steps for solving $x^2 + 4x - 12 = 0$ in the right order.

- ☐ $(x + 2)^2 = 16$
☐ $x + 2 = 4$ or $x + 2 = -4$
☐ $x = 2$ or $x = -6$
☐ $x^2 + 4x = 12$
☐ $x^2 + 4x + 4 = 16$

7.

Solve $3x^2 + 12x + 6 = 0$ by completing the square. Show each step and give the exact roots.

A large grid of 20 columns and 20 rows, intended for the student to show their work in completing the square.

EXAM TASK

Not quite a square

Level 3 · basic

PUPP style · 3 points

The expression $x^2 - 6x + 11$ is given.

a) Show that $x^2 - 6x + 11 = (x - 3)^2 + 2$.

b) Solve the equation $x^2 - 6x + 11 = 6$.

c) Explain why the equation $x^2 - 6x + 11 = 0$ has no real roots.

